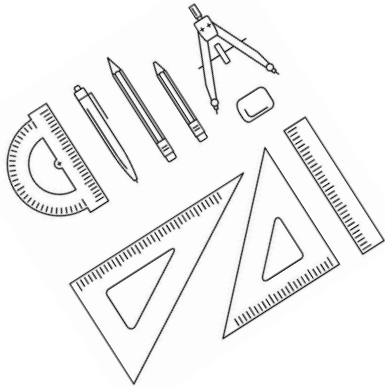


PROJECTION

Course Code: CSE200

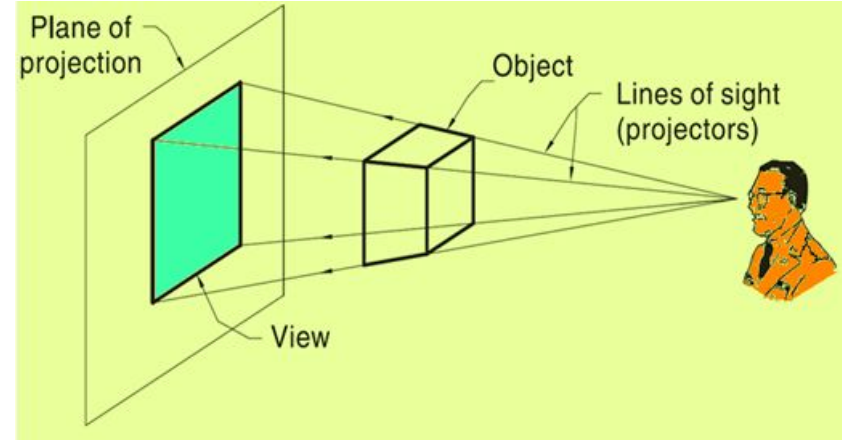


Projection

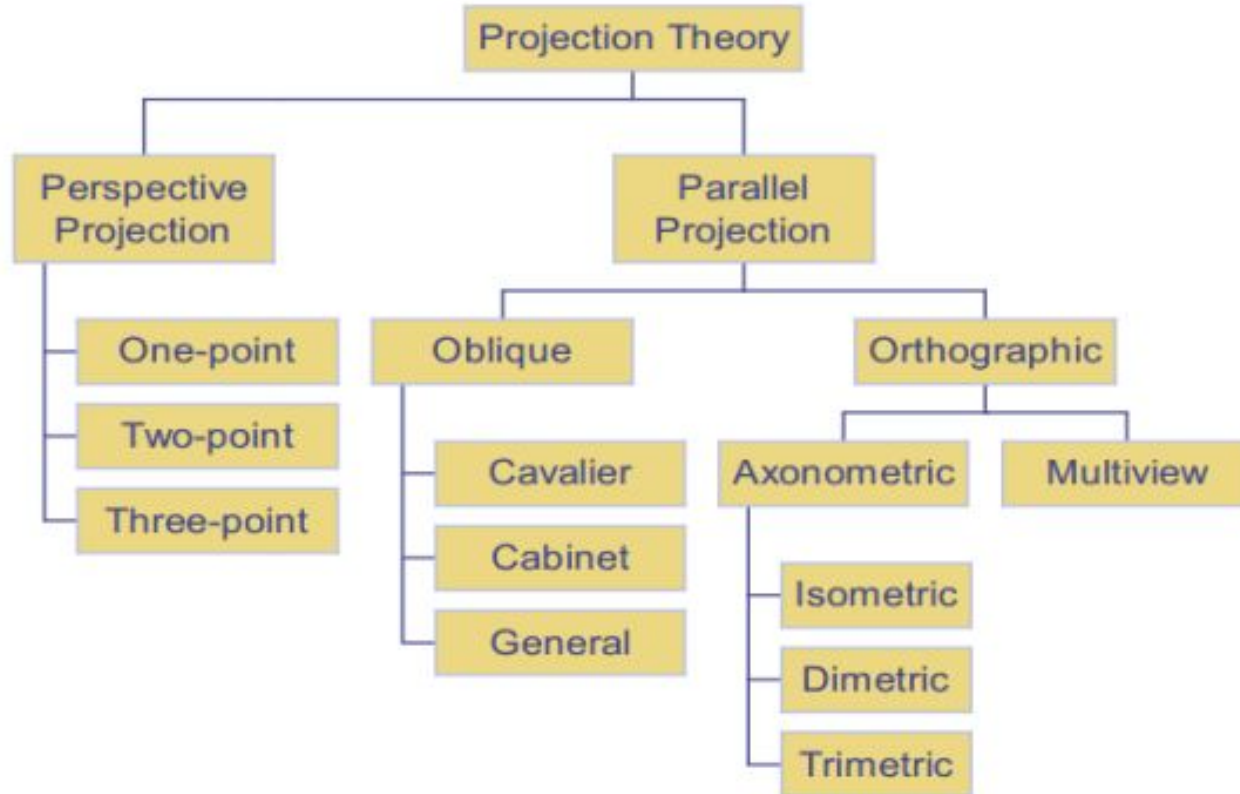
Projection refers to the process of representing a three-dimensional object on a two-dimensional plane or surface.

A **line of projection** (or **projectors**) is an imaginary line that extends from a point on the object to the plane of projection.

The **station point** is the location of the observer's eye. It is the point from which the lines of projection originate



Classification of Projection

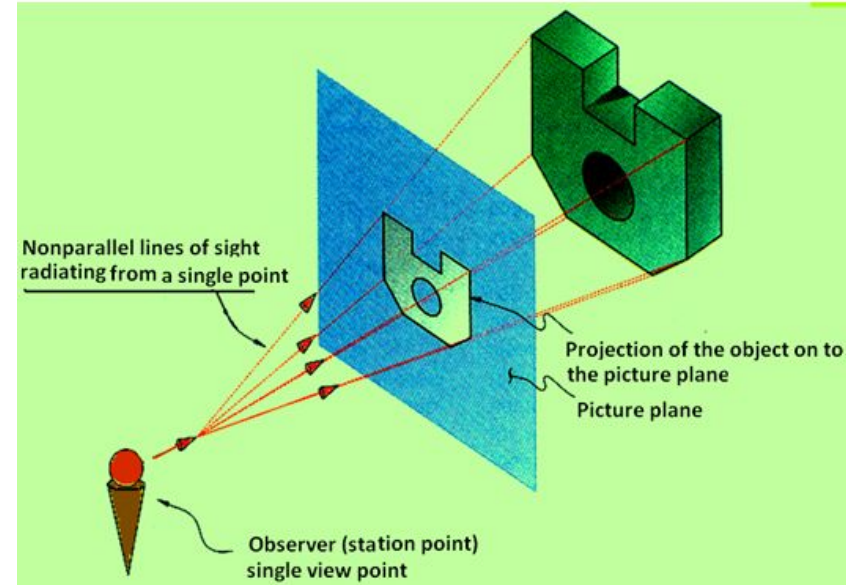


Types of Projection (Perspective Projection)

Perspective Projection all lines of sight start at a single point.

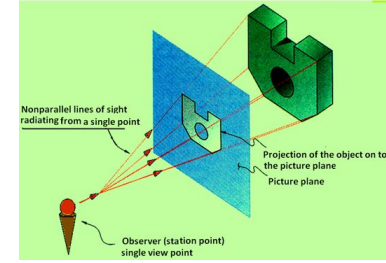
Types of Projection (Perspective Projection)

Perspective Projection all lines of sight start at a single point.



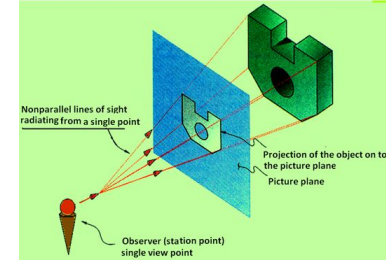
Types of Projection (Perspective Projection)

Perspective Projection all lines of sight start at a single point.



Types of Projection (Parallel Projection)

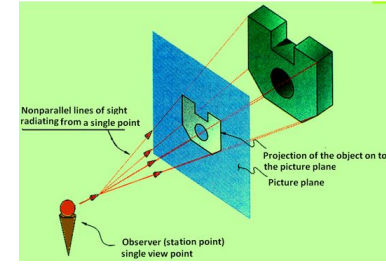
Perspective Projection all lines of sight start at a single point.



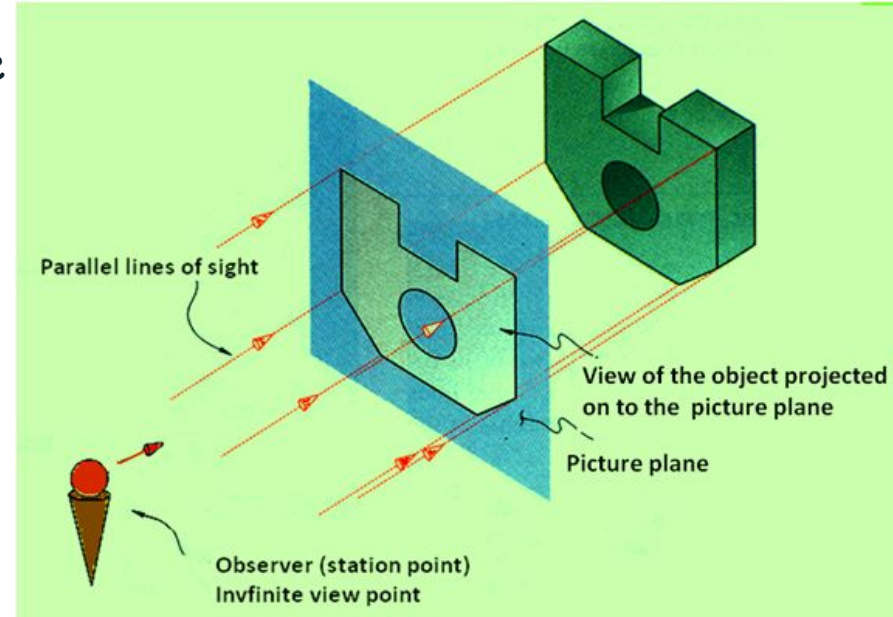
In **parallel projection**, all lines of sight are parallel.

Types of Projection (Parallel Projection)

Perspective Projection all lines of sight start at a single point.



In **parallel projection**, all lines of sight are parallel.



Orthographic Projection (Parallel Projection)

Orthographic projection is a parallel projection technique in which the plane of projection is perpendicular to the parallel line of sight.

Orthographic Projection (Parallel Projection)

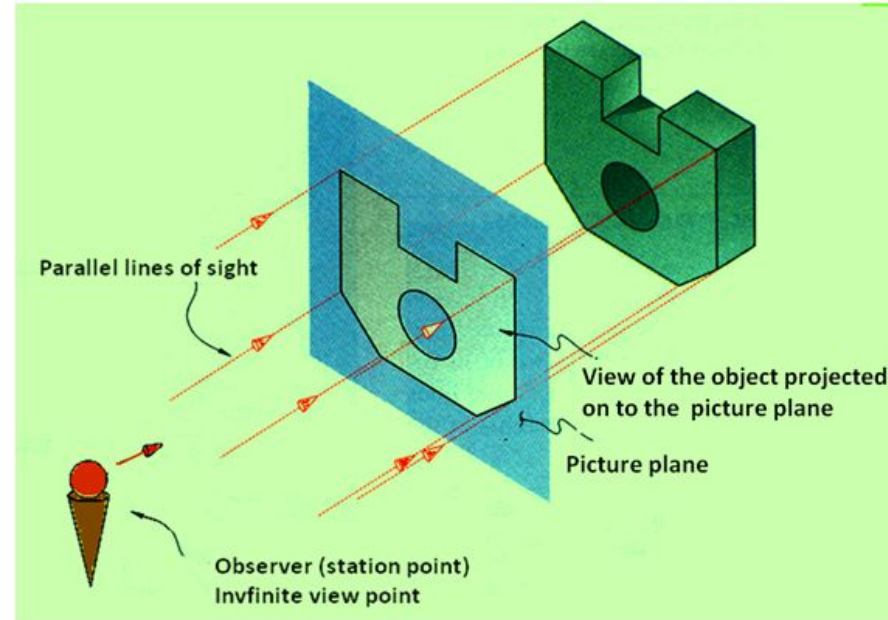
Orthographic projection is a parallel projection technique in which the plane of projection is perpendicular to the parallel line of sight.

Therefore we get the projection as **exact size of the object**.

Orthographic Projection (Parallel Projection)

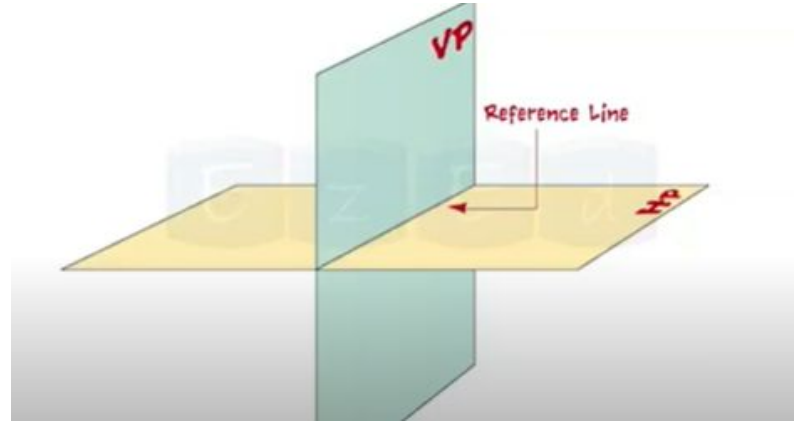
Orthographic projection is a parallel projection technique in which the plane of projection is perpendicular to the parallel line of sight.

Therefore we get the projection as **exact size of the object**.



Reference Frame

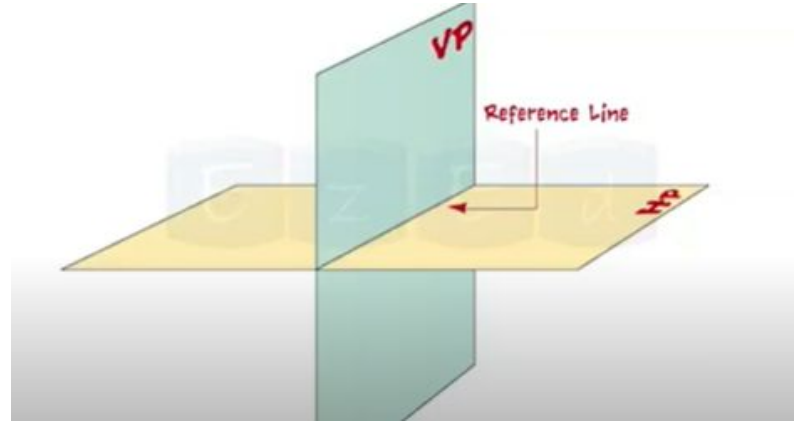
Vertical Plane (VP): This plane is vertical and perpendicular to the ground plane. Also known as Frontal Reference Frame.



Reference Frame

Vertical Plane (VP): This plane is vertical and perpendicular to the ground plane. Also known as Frontal Reference Frame.

Horizontal Plane (HP): This plane is horizontal and perpendicular to the vertical plane.

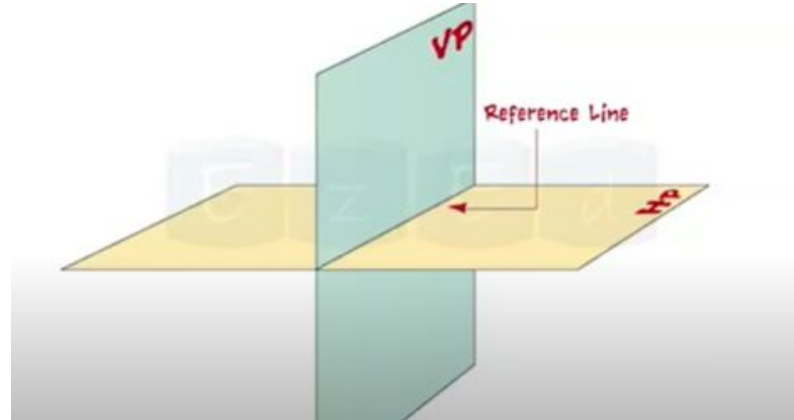


Reference Frame

Vertical Plane (VP): This plane is vertical and perpendicular to the ground plane. Also known as Frontal Reference Frame.

Horizontal Plane (HP): This plane is horizontal and perpendicular to the vertical plane.

Reference Line: Intersecting line of HP and VP.



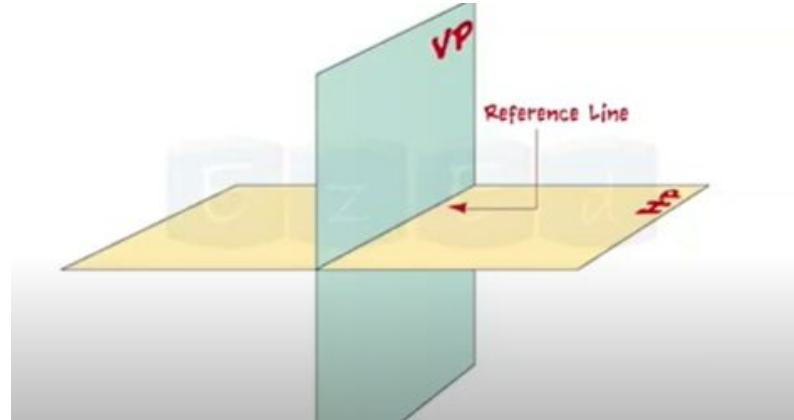
Reference Frame

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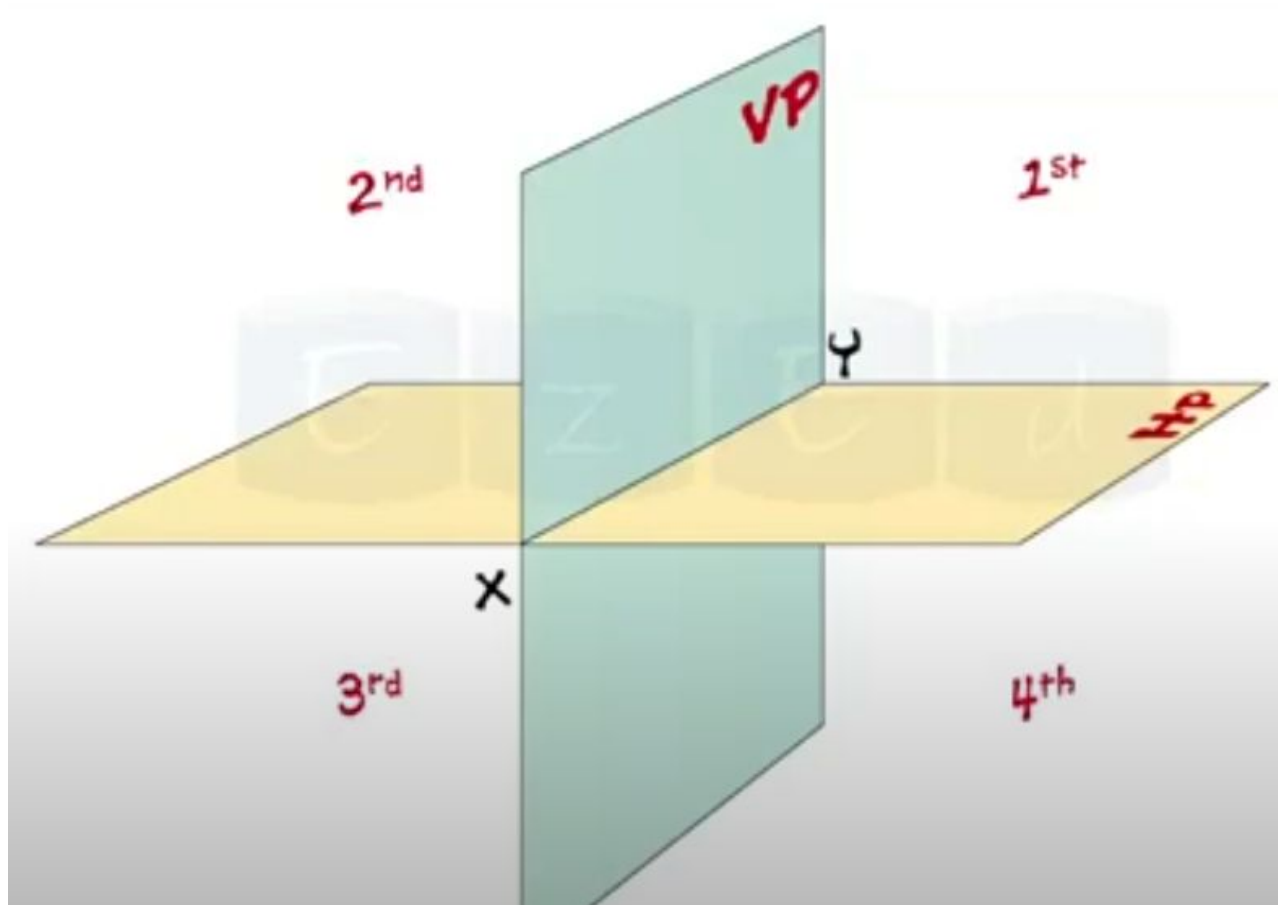
Horizontal Plane (HP): This plane is horizontal and perpendicular to the vertical plane.

Reference Line: Intersecting line of HP and VP.

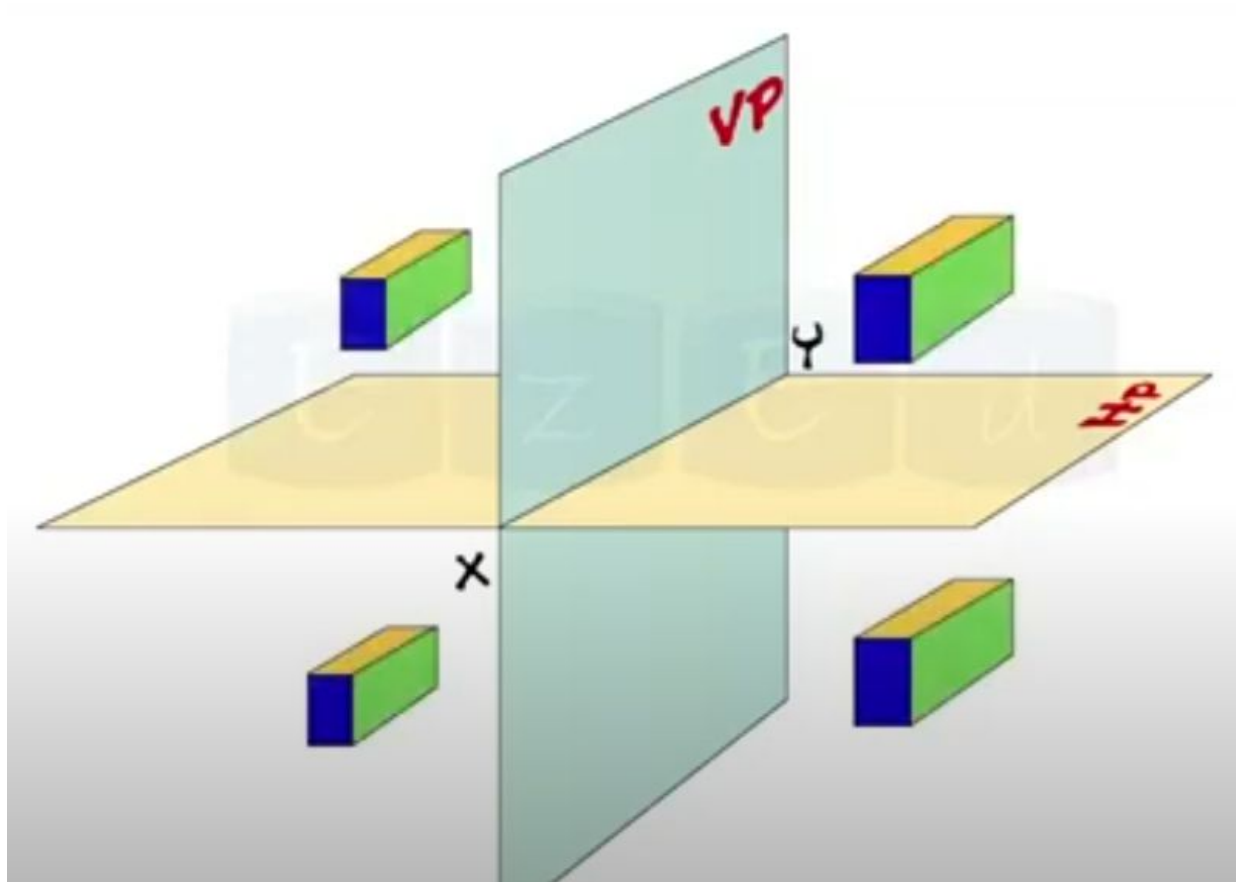
Profile Plane (PP): This plane is perpendicular to both the horizontal plane (HP) and the vertical plane (VP) in orthographic projection. It is used to project the side view of an object. Also known as **Auxiliary Vertical Plane (AVP)**.



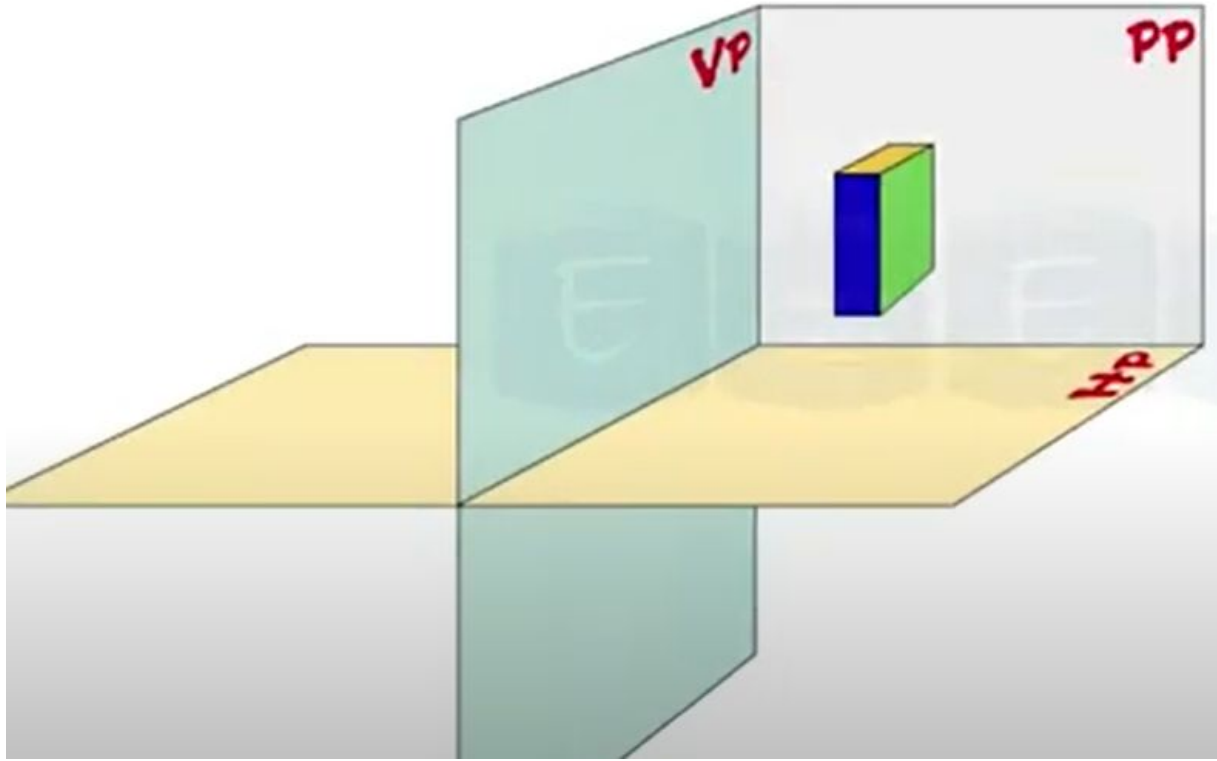
Reference Frame



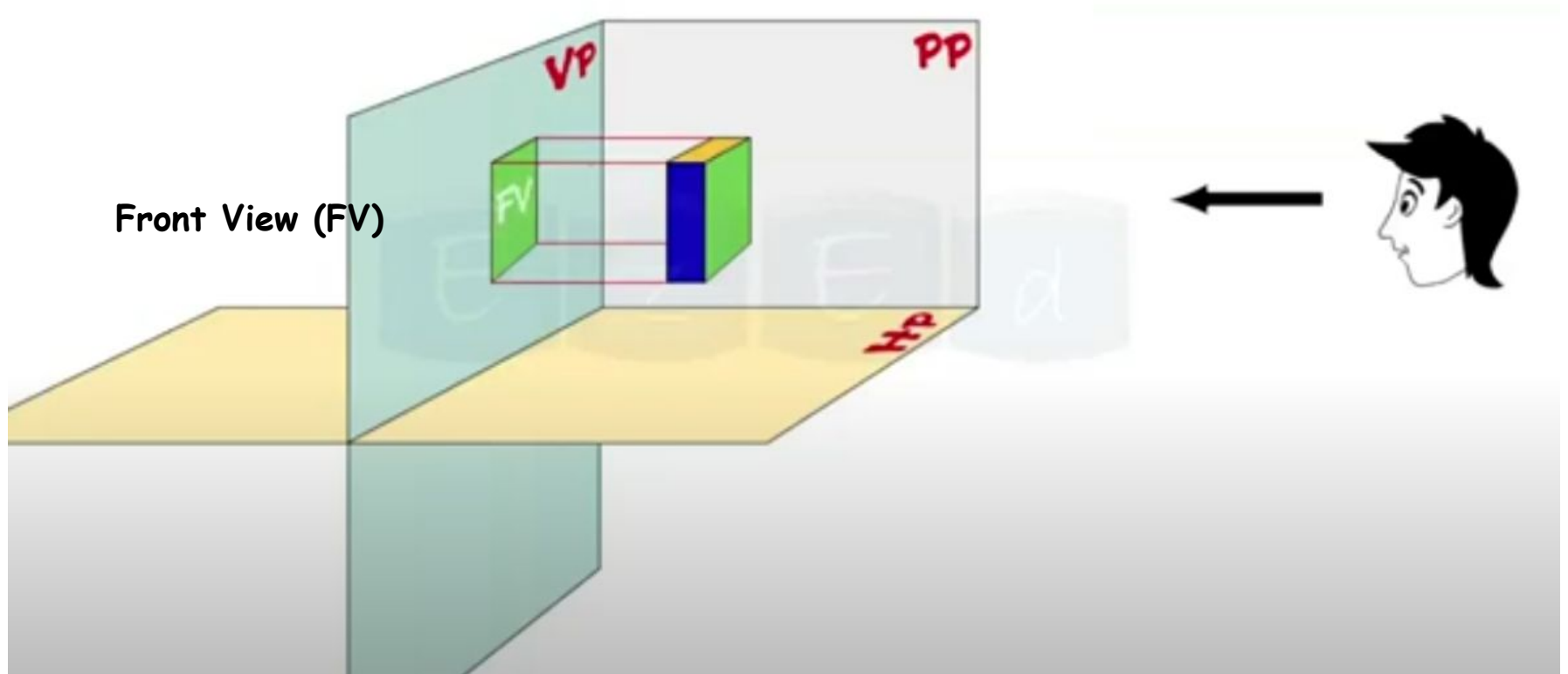
Reference Frame



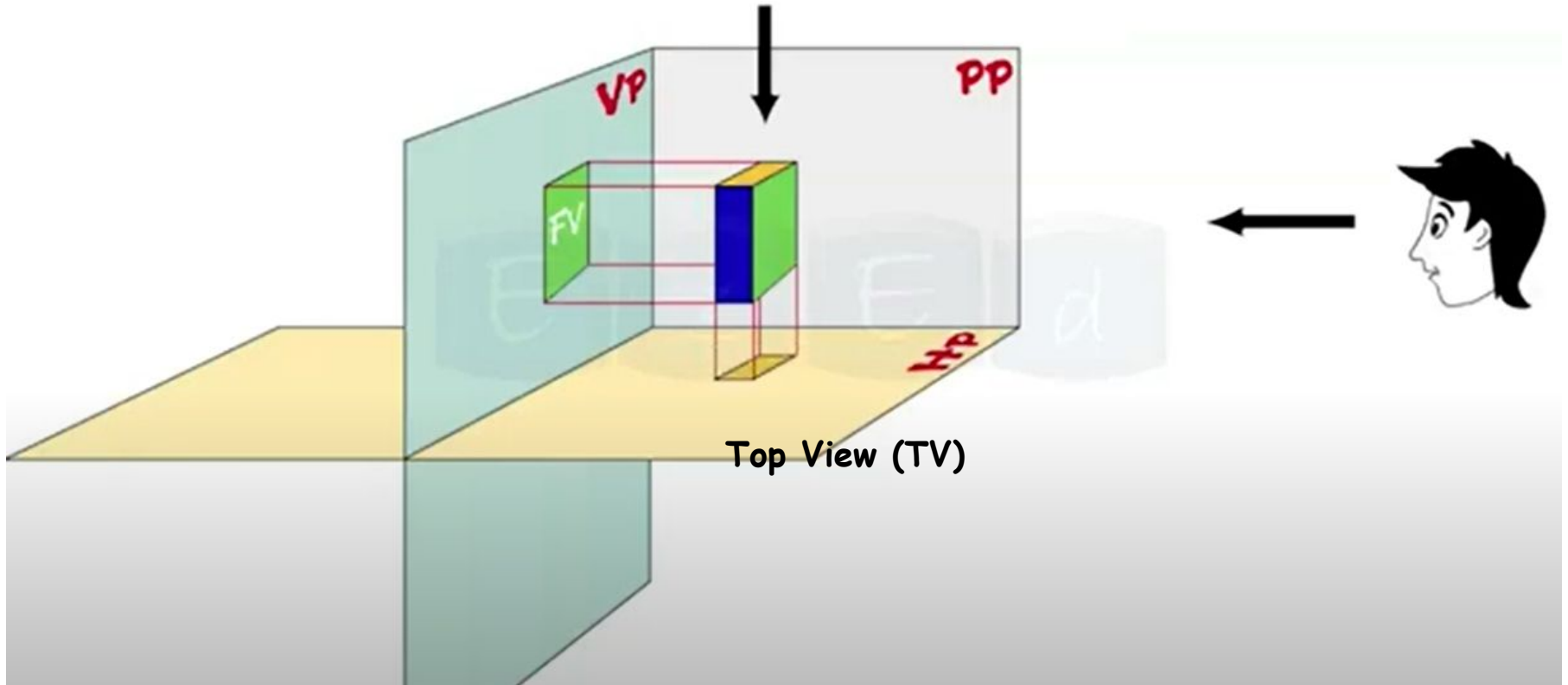
Method of Projection



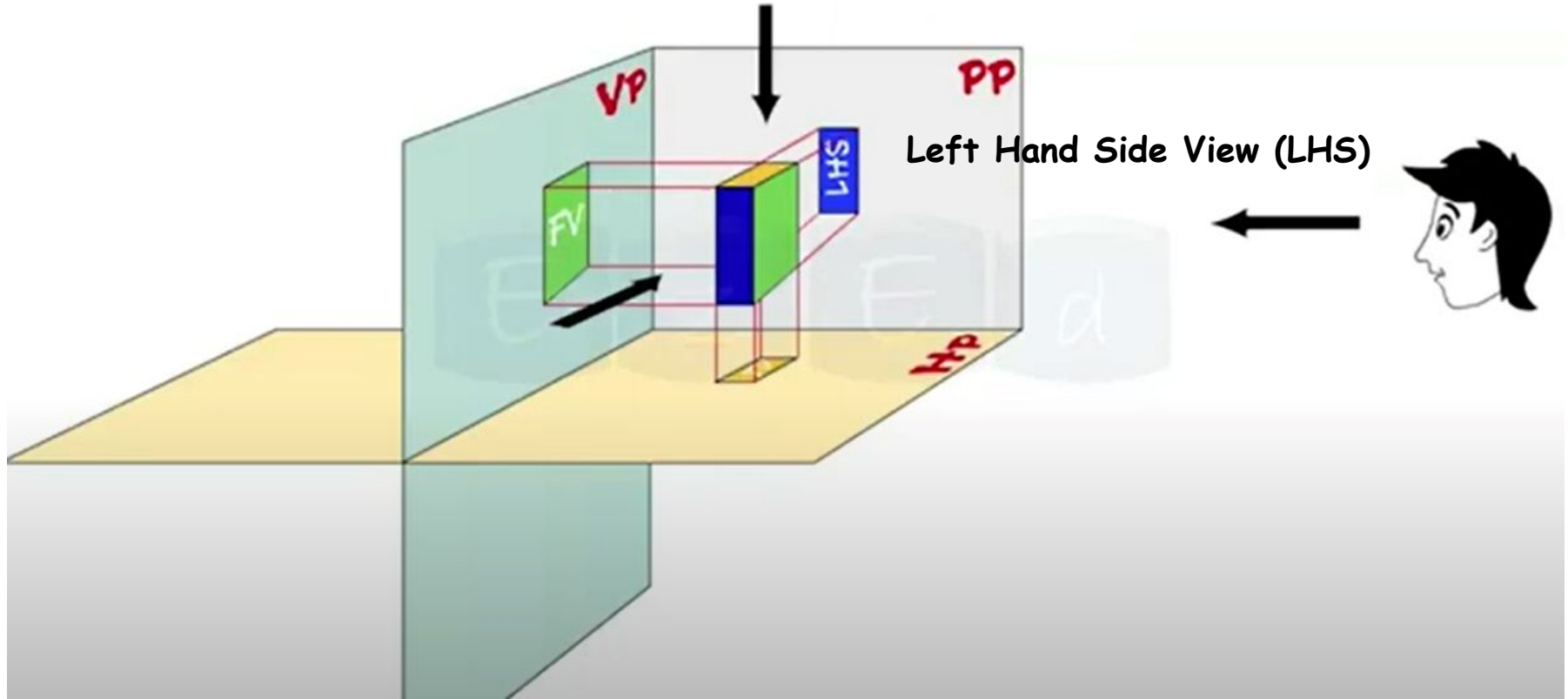
Method of Projection



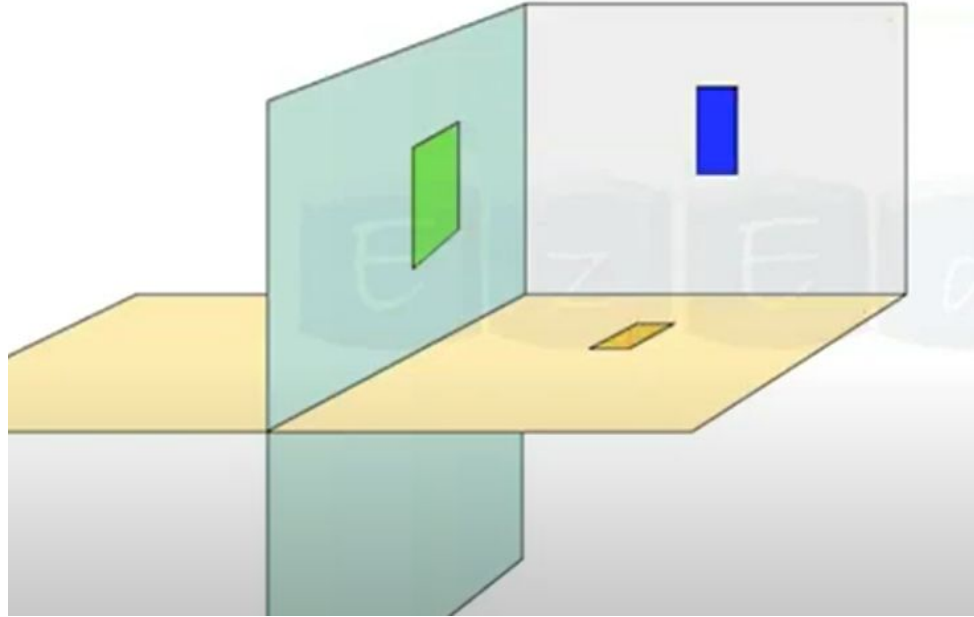
Method of Projection



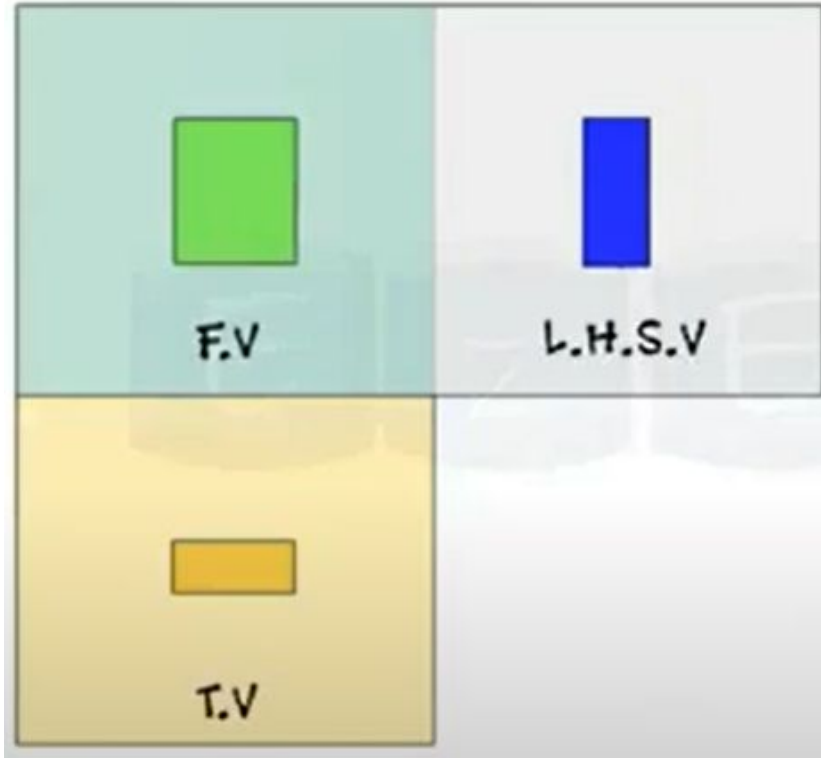
Method of Projection



Method of Projection



Method of Projection

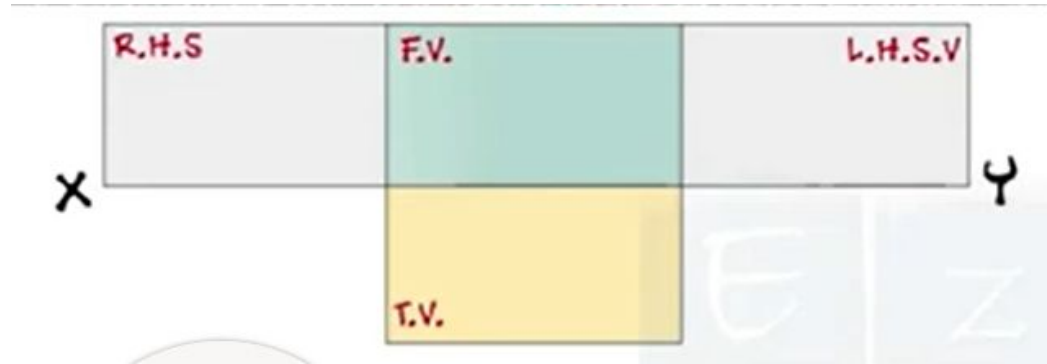


First Angle Method of Projection

In industries the term Front View is referred to as **Elevation** and Top View is referred to as **Plan**,

Method of Projection

Right Hand Side (RHS) is displayed on the left.



First Angle Method of Projection

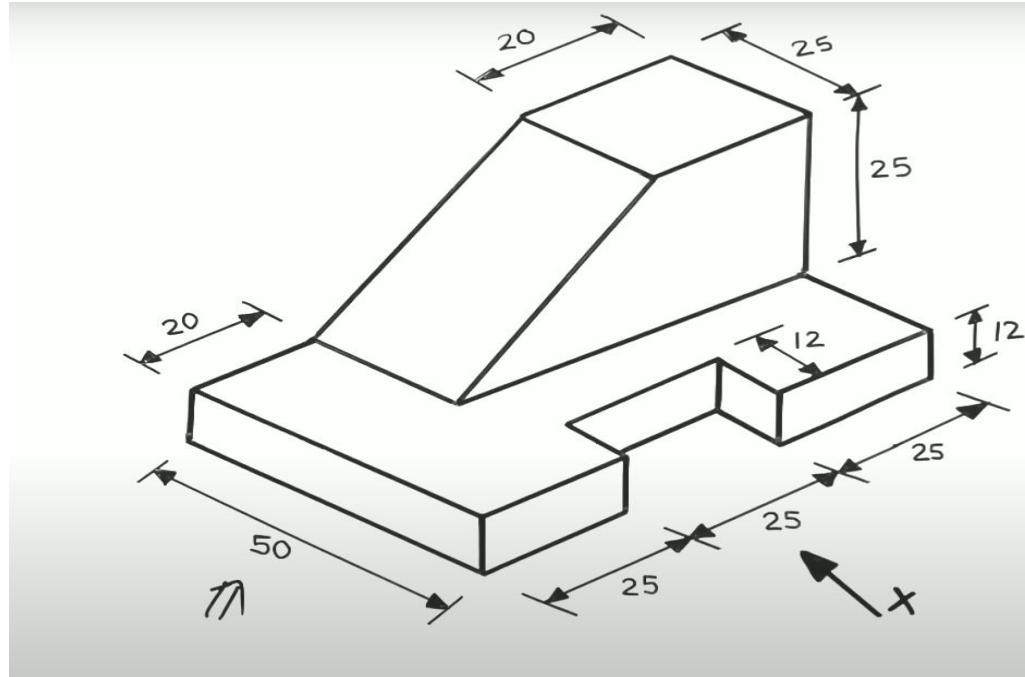
In first angle projection, the object is conceptually placed in the first quadrant of a three-dimensional space. The viewer is positioned between the object and the projection planes. This means that the projection of the object onto the planes is a result of the viewer's perspective.

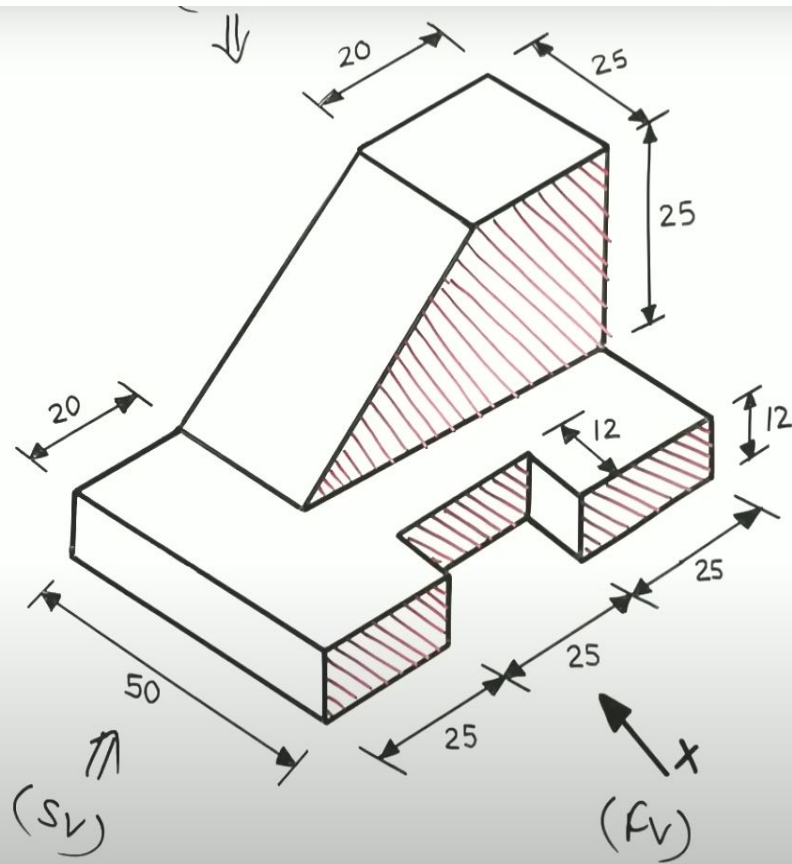
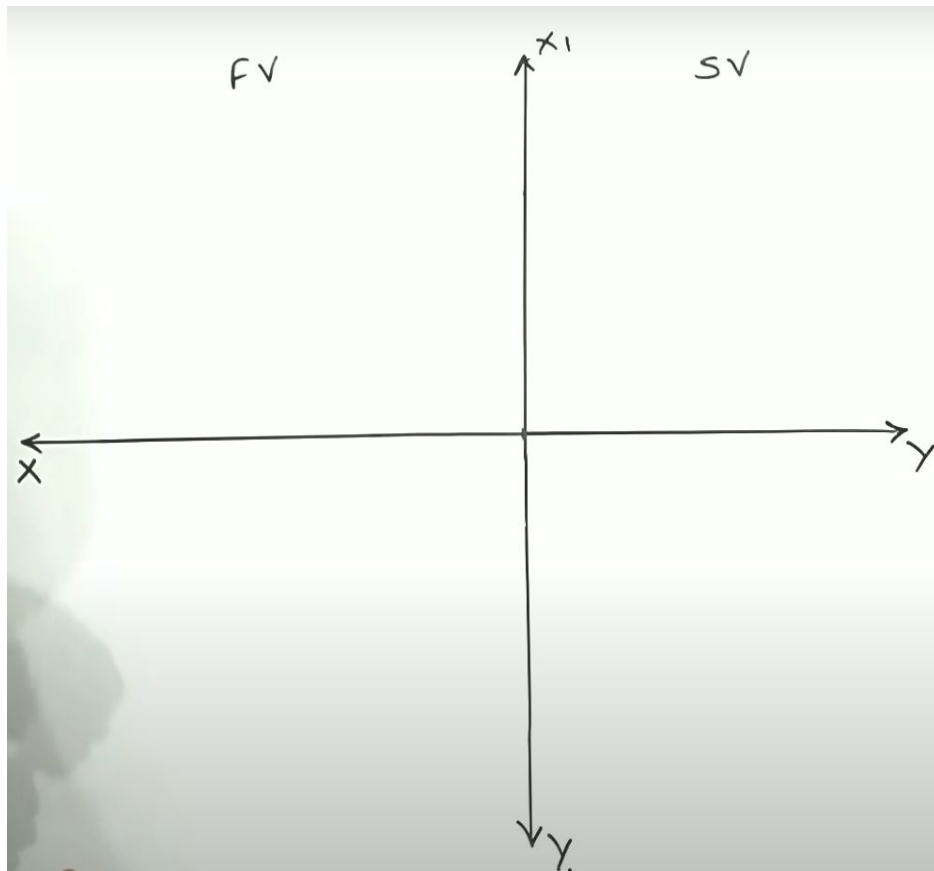
Here's a breakdown of the method:

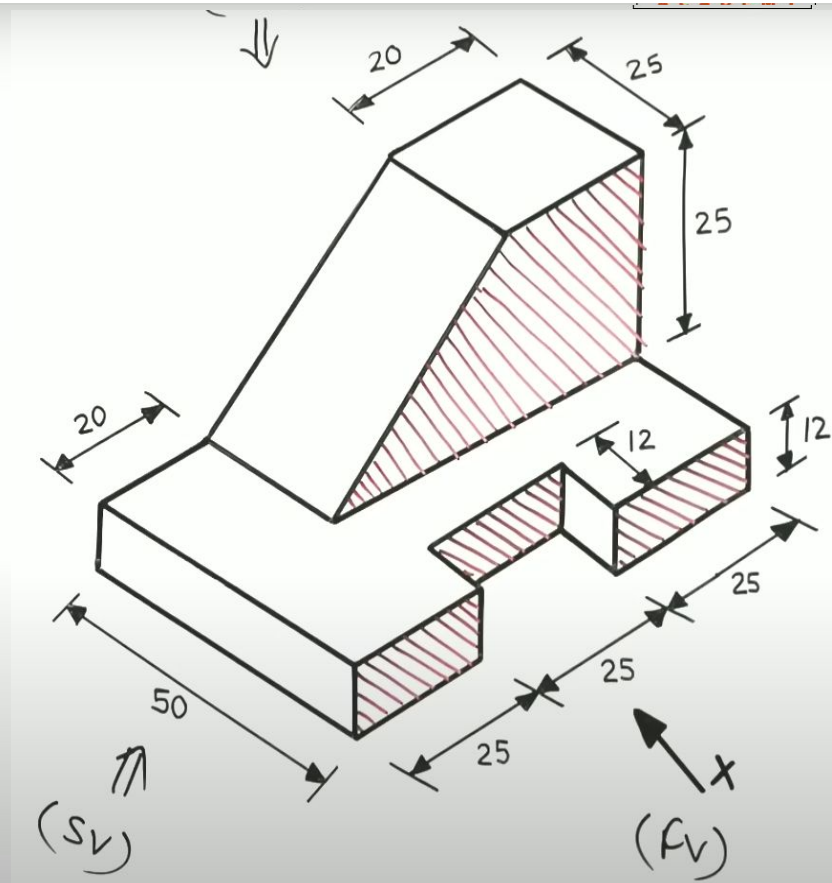
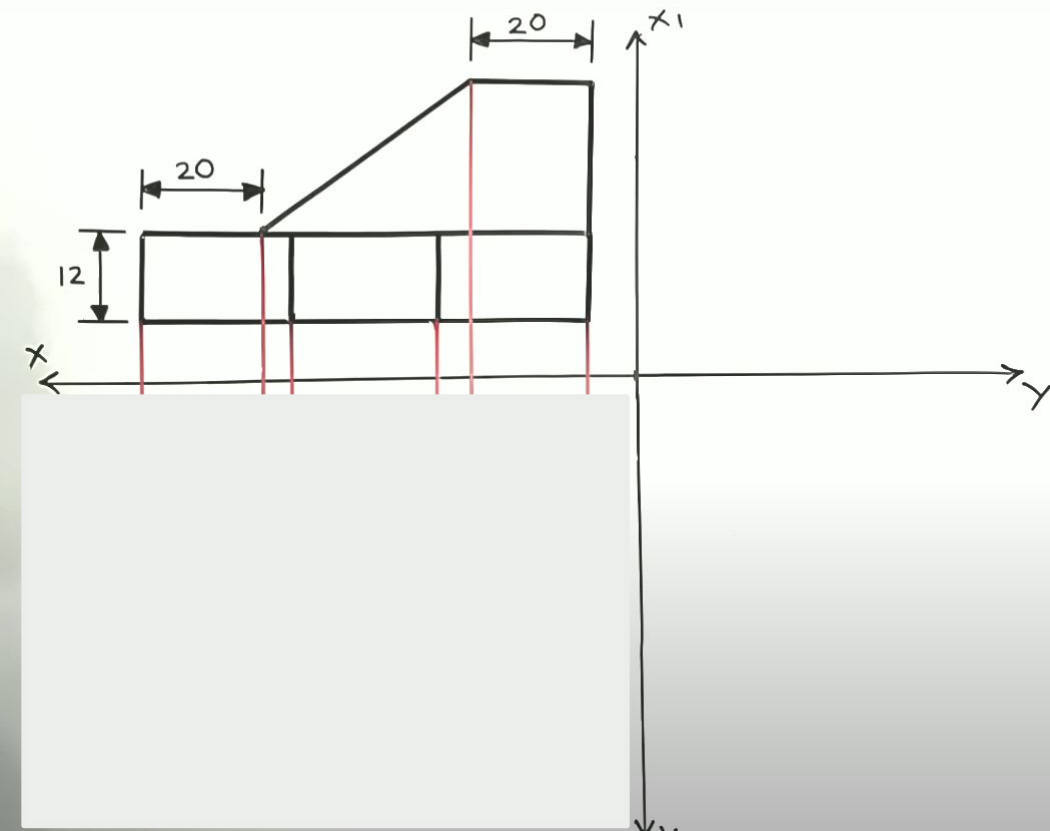
1. **Object Placement:** The object is imagined to be placed in the first quadrant.
2. **Projection Planes:** The object is projected onto six planes: front, top, right side, left side, bottom, and back.
3. **View Arrangement:** The views are arranged as follows:
 - **Front View:** At the bottom of the drawing sheet.
 - **Top View:** Above the front view.
 - **Right Side View:** To the left of the front view.
 - **Left Side View:** To the right of the front view.
 - **Bottom View:** Below the front view.
 - **Back View:** To the right of the right side view.

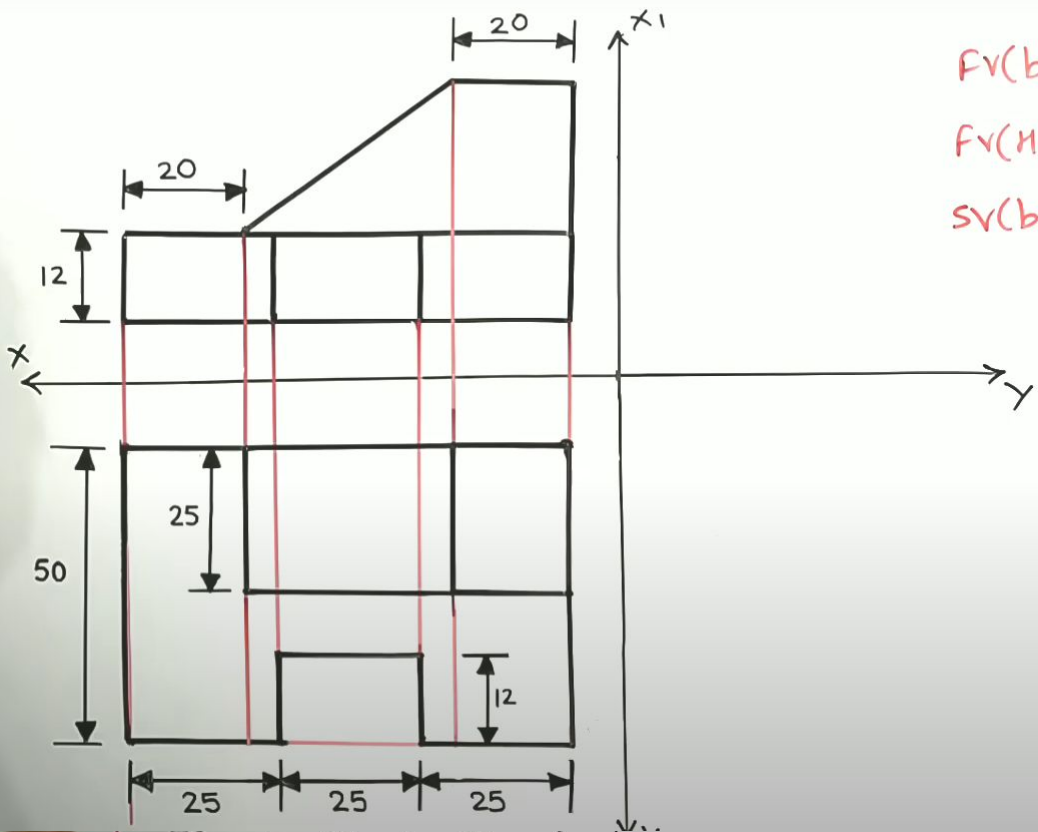
Practice Problem 1

- Draw the front view, Top view and Side view of the given figure.









$$FV(bs) = TV(bs)$$

$$FV(ht) = SV(ht)$$

$$SV(bs) = TV(ht)$$

